PROGRAM:

AIES-Exp_3.py - C:/Users/ABBIRAMY/OneDrive/Documents/AIES_LAB/AIES-Exp_3.py (3.14.2)

File Edit Format Run Options Window Help

```
import heapq
graph = {
    'S': {'A': 1, 'B': 4},
    'A': {'B': 2, 'C': 5, 'D': 12},
    'B': {'C': 2},
    'C': {'G': 3},
    'D': {'G': 3},
    'G': {}
}
def ucs(graph, start, goal):
    pq = [(0, start, [start])]    # (cost so far, node, path)
    visited = set()
    while pq:
        cost, node, path = heapq.heappop(pq)
        if node == goal:
            return path, cost
        if node in visited:
            continue
        visited.add(node)
        for nxt, w in graph[node].items():
            if nxt not in visited:
                heapq.heappush(pq, (cost + w, nxt, path + [nxt]))
    return None, float('inf')
path, cost = ucs(graph, 'S', 'G')
print("Least-cost path:", path)
print("Total cost     :", cost)
```

OUTPUT:

IDLE Shell 3.14.2

File Edit Shell Debug Options Window Help

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/ABBIRAMY/OneDrive/Documents/AIES_LAB/AIES-Exp_3.py =====
Least-cost path: ['S', 'A', 'B', 'C', 'G']
Total cost     : 8
>>>
```